

I was primarily focused on dataset acquisition and preprocessing. I downloaded the BBC News dataset, cleaned missing values, normalized text, and implemented tokenization and lemmatization. Through this process I gained a deeper understanding of text preparation and its impact on downstream machine learning performance. One challenge was ensuring consistency across thousands of articles while maintaining computational efficiency in Google Colab. The experience strengthened my Python and NLP preprocessing skills.

I then focused on TF-IDF analysis, feature engineering, and visualization. I explored how different vectorization parameters affected model performance and generated visualizations that highlighted category-specific terminology. This task helped me understand how numerical representations of language influence classification algorithms. I also learned how to communicate technical findings through meaningful visualizations.

My responsibilities included sentiment analysis, POS tagging, dependency parsing, and named entity recognition. I developed functions that extracted linguistic information from news articles and interpreted patterns across categories. One of the most valuable lessons was seeing how syntactic and semantic features can provide additional insight beyond traditional machine learning metrics.

I focused on machine learning model development and evaluation. I implemented Logistic Regression, Random Forest, and Naive Bayes classifiers, compared their performance, and documented the results. Through this work I improved my understanding of model selection, evaluation metrics, and performance optimization.

This project demonstrated the importance of integrating multiple NLP techniques into a complete intelligence system.